TUTORIAL ASSIGNMENT (15%)

Question 1 [5 marks]

Answer the following questions for the array shown below.

$$arr = \begin{bmatrix} 3.1 & 0.5 & 2.2 & -3.8 & 6.0 \\ -0.9 & -1.3 & -8.8 & 1.1 & 4.3 \\ 1.5 & 0.2 & 0.7 & -0.6 & 1.3 \\ 1.5 & 5.1 & 0.0 & 1.4 & -2.1 \end{bmatrix}$$

i. What is the size of arr?
ii. What is the value of arr (3, 3)?
iii. What is the value of arr (12)?
iv. What is the size and value of arr (:, 1:2:4)?
v. What is the size and value of arr ([1 4], [end-1 end])?

Question 2 [5 marks]

Assume that array z is defined as shown, determine the contents of the following subarrays:

$$z = \begin{bmatrix} 1.1 & -3.2 & 3.4 & 0.6 \\ -0.8 & 1.3 & -0.4 & 3.1 \\ -2.1 & 0.6 & 2.2 & 0.0 \\ 1.1 & 0.1 & 11.1 & -0.9 \end{bmatrix}$$

i. z (2, :)
ii. z (:, end)
iii. z (1 : 2, 2 : end)
iv. z (4 : end)
v. z ([1 3], 2)

Question 3 [5 marks]

The distance between two points (x_1, y_1, z_1) and (x_2, y_2, z_2) in a three-dimensional Cartesian coordinate system is given by the equation

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

Write a program that

a) prompts the user to enter the values of x_1, y_1, z_1, x_2, y_2, and z_2;
b) calculates and displays the distance between any two points (x_1, y_1, z_1) and (x_2, y_2, z_2) specified by the user.